AQA A-Level Further Mathematics (7367)

Question Paper

Further Calculus

E7.2 - Area of Surface of Rev

(A-Level)

Name: ______________________

Class: ______________________

Date: ______________________

Time:	102 min.
Marks:	84 marks
Comments:	

Q1.

(a) (i) Show that

$$\frac{\mathrm{d}}{\mathrm{d}u}\left(2u\sqrt{1+4u^2}+\sinh^{-1}2u\right)=k\sqrt{1+4u^2}$$

where k is an integer.

(4)

(ii) Hence show that

$$\int_0^1 \sqrt{1+4u^2}\,\mathrm{d}u = p\sqrt{5}+q\sinh^{-1}2$$

where p and q are rational numbers.

(2)

(b) The arc of the curve with equation $y = \frac{1}{2}\cos 4x$ between the points where $x = 0$ and $x = \frac{\pi}{8}$ is rotated through 2π radians about the x-axis.

(i) Show that the area S of the curved surface formed is given by

$$S = \pi\int_0^{\frac{\pi}{8}} \cos 4x\sqrt{1+4\sin^2 4x}\,\mathrm{d}x$$

(2)

(ii) Use the substitution $u = \sin 4x$ to find the exact value of S.

(4)

(Total 12 marks)

Q2.

(a) Show that

$$\frac{1}{4}(\cosh 4x + 2\cosh 2x + 1) = \cosh^2 x\cosh 2x$$

(3)

(b) Show that, if $y = \cosh^2 x$, then

$$1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 = \cosh^2 2x$$

(3)

(c) The arc of the curve $y = \cosh^2 x$ between the points where $x = 0$ and $x = \ln 2$ is rotated through 2π radians about the x-axis. Show that the area S of the curved surface formed is given by

$$S = \frac{\pi}{256}\,(a \ln 2 + b)$$

where a and b are integers.

(7)
(Total 13 marks)

Q3.

(a) The arc of the curve $y^2 = x^2 + 8$ between the points where $x = 0$ and $x = 6$ is rotated through 2π radians about the x-axis. Show that the area S of the curved surface formed is given by

$$S = 2\sqrt{2}\pi\int_0^6 \sqrt{x^2+4}\,\mathrm{d}x$$

(5)

(b) By means of the substitution $x = 2\sinh\theta$, show that

$$S = \pi\left(24\sqrt{5} + 4\sqrt{2}\sinh^{-1}3\right)$$

(8)
(Total 13 marks)

Q4.

A curve C is given parametrically by the equations

$$x = \frac{1}{2}\cosh 2t, \qquad y = 2\sinh t$$

(a) Express

$$\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2$$

in terms of $\cosh t$.

(6)

(b) The arc of C from $t = 0$ to $t = 1$ is rotated through 2π radians about the x-axis.

(i) Show that S, the area of the curved surface generated, is given by

$$S = 8\pi\int_0^1 \sinh t\cosh^2 t\;\mathrm{d}t$$

(2)

(ii) Find the exact value of S.

(2)
(Total 10 marks)

Q5.

(a) Use the definition $\cosh x = \frac{1}{2}\left(e^x + e^{-x}\right)$ to show that $\cosh 2x = 2\cosh^2 x - 1$.

(2)

(b) (i) The arc of the curve $y = \cosh x$ between $x = 0$ and $x = \ln a$ is rotated through 2π radians about the x-axis. Show that S, the surface area generated, is given by

$$S = 2\pi\int_0^{\ln a} \cosh^2 x\,dx$$

(3)

(ii) Hence show that

$$S = \pi\left(\ln a + \frac{a^4 - 1}{4a^2}\right)$$

(5)

(Total 10 marks)

Q6.

(a) Given that $y = \operatorname{sech} t$, show that:

(i) $\frac{dy}{dt} = -\operatorname{sech} t \tanh t$

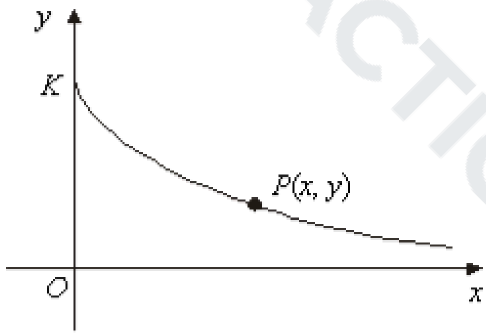
(3)

(ii) $\left(\frac{dy}{dt}\right)^2 = \operatorname{sech}^2 t - \operatorname{sech}^4 t.$

(2)

(b) The diagram shows a sketch of part of the curve given parametrically by

$x = t - \tanh t$ $\quad y = \operatorname{sech} t$



The curve meets the y-axis at the point K, and $P(x, y)$ is a general point on the curve. The arc length KP is denoted by s. Show that:

(i) $$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \tanh^2 t;$$

(4)

(ii) $s = \ln \cosh t$;

(3)

(iii) $y = e^{-s}$.

(2)

(c) The arc KP is rotated through 2π radians about the x-axis. Show that the surface area generated is $2\pi\,(1 - e^{-s})$

(4)

(Total 18 marks)

Q7.

A curve has parametric equations

$$x = t - \frac{1}{3}t^3, \quad y = t^2$$

(a) Show that

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (1+t^2)^2$$

(3)

(b) The arc of the curve between $t = 1$ and $t = 2$ is rotated through 2π radians about x-axis.

Show that S, the surface area generated, is given by $S = k\pi$, where k is a rational number to be found.

(5)

(Total 8 marks)